Progress report 1: base rate in the prompt, or in code?

2026-09-28, 19:45 EDT · autonomous v2 session · Haiku 4.5 still running (rate-limited), so every count below covers 8 of 9 model configurations.

Where things stand

> **Candidate headline finding.** Put a known base rate in an LLM's prompt ("about 22% of cases like these are yes"). For 7 of the 8 configurations tested, the resulting probabilities score *worse* (higher Brier) than simply predicting that 22% for every case. They also score worse than applying the same number in code as a one-parameter shift of the model's untold scores. The one exception is Gemini 3.8 Flash, the strongest ranker here.
>
> - This pattern emerged post hoc in E1 (8 tasks). It then held in **E1-R**, a preregistered replication on 5 fresh September-2026 task samples that no model or analysis had touched.
> - Every predicted sign was confirmed after Holm correction, and all 3 seeds agree.
> - Haiku's replication is still pending.

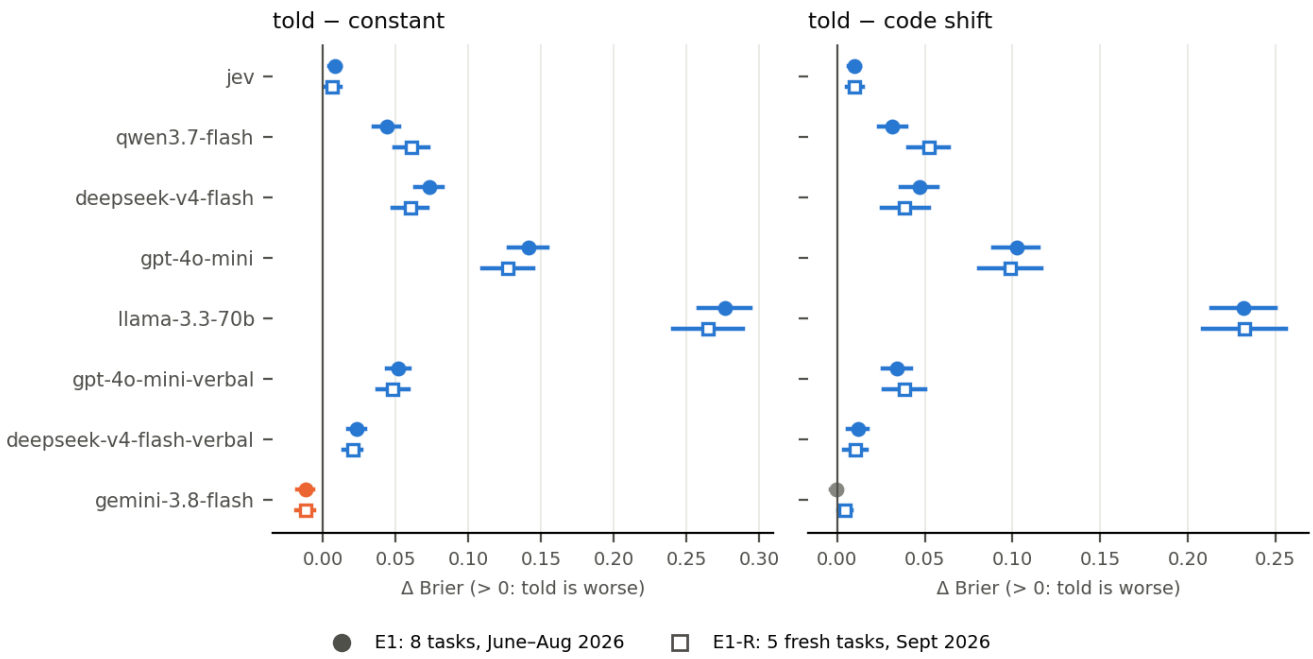
The originally preregistered E1 primary criterion (told vs shift significant on ≥5/8 tasks per model) was met by only 2 configurations, GPT-4o-mini and Llama-3.3-70B. The v1 claim "5/8 for Jev" did not replicate on the leak-repaired data: Jev reached 3/8.

What was done since the start of this session

- **Validity fixes.**
  - A single-feature leak screen covered every task and feature.
  - Repaired leaks: shelter (`sex`, `intake_condition` dropped; alone, `sex` gave CV AUROC 0.78), GitHub (the `draft` state at closing), and crimes (police-initiated offence types dropped).
- **New baselines, trained on the same 100 labels that give the stated rate.** A constant, Platt(100), and logistic regression LR(100).
- **9 model configurations, 7 providers, each pinned to one provider.**
  - First-token logprobs: Jev (native probabilities), Qwen3.7-Flash, DeepSeek-V4-Flash, GPT-4o-mini, Llama-3.3-70B.
  - Stated probabilities: Gemini 3.8 Flash, Claude Haiku 4.5, and GPT-4o-mini and DeepSeek asked the same way (same models, different elicitation).
  - GPT-4o-mini and Llama have 2023 training cutoffs, and E1-R data is from September 2026, after every model's cutoff. That addresses the memorisation concern.
- **Preregistrations** (RESEARCH_AGENDA.md §2 and §3) were committed before each data collection, and every deviation is logged in AUTONOMOUS_LOG.md.

Results (Brier differences; positive = the first-named arm is worse)

Pooled over tasks per model (mean of per-task, seed-averaged differences; bootstrap within task, B=5000). **Bold** = significant after Holm across models. Each model has 3 seeds; each seed is a different 100-label estimate of the base rate.



E1-R (preregistered confirmatory; 5 fresh tasks × 3 seeds × 300 test cases)

Model	told − constant	told − code shift	code shift − constant	Platt(100) − code shift
jev	**+0.006 [+0.001, +0.012]**	**+0.010 [+0.005, +0.015]**	**-0.004 [-0.006, -0.001]**	+0.000 [-0.003, +0.003]
qwen3.7-flash	**+0.061 [+0.049, +0.073]**	**+0.052 [+0.041, +0.064]**	**+0.009 [+0.003, +0.015]**	**-0.011 [-0.016, -0.006]**
deepseek-v4-flash	**+0.060 [+0.048, +0.072]**	**+0.039 [+0.025, +0.052]**	**+0.021 [+0.014, +0.029]**	**-0.023 [-0.030, -0.016]**
gpt-4o-mini	**+0.127 [+0.110, +0.144]**	**+0.098 [+0.081, +0.117]**	**+0.028 [+0.019, +0.038]**	**-0.028 [-0.038, -0.019]**
llama-3.3-70b	**+0.265 [+0.241, +0.289]**	**+0.232 [+0.209, +0.256]**	**+0.032 [+0.022, +0.042]**	**-0.032 [-0.041, -0.022]**
gpt-4o-mini-verbal	**+0.048 [+0.037, +0.059]**	**+0.038 [+0.027, +0.050]**	**+0.010 [+0.005, +0.015]**	**-0.009 [-0.014, -0.004]**
deepseek-v4-flash-verbal	**+0.020 [+0.014, +0.027]**	**+0.010 [+0.004, +0.017]**	**+0.010 [+0.004, +0.016]**	**-0.008 [-0.014, -0.002]**
gemini-3.8-flash	**-0.012 [-0.019, -0.006]**	**+0.004 [+0.000, +0.008]**	**-0.017 [-0.024, -0.010]**	-0.001 [-0.003, +0.001]

E1 (8 tasks; told − shift was the preregistered primary, the other columns are post hoc)

Model	told − constant	told − code shift	code shift − constant	Platt(100) − code shift
jev	**+0.008 [+0.004, +0.012]**	**+0.010 [+0.007, +0.013]**	-0.001 [-0.004, +0.001]	-0.000 [-0.003, +0.002]
qwen3.7-flash	**+0.044 [+0.035, +0.052]**	**+0.032 [+0.024, +0.040]**	**+0.012 [+0.007, +0.018]**	**-0.014 [-0.018, -0.010]**
deepseek-v4-flash	**+0.073 [+0.063, +0.083]**	**+0.047 [+0.036, +0.057]**	**+0.026 [+0.019, +0.033]**	**-0.026 [-0.032, -0.019]**
gpt-4o-mini	**+0.141 [+0.128, +0.154]**	**+0.102 [+0.089, +0.115]**	**+0.039 [+0.031, +0.047]**	**-0.036 [-0.044, -0.028]**
llama-3.3-70b	**+0.276 [+0.258, +0.294]**	**+0.232 [+0.213, +0.250]**	**+0.044 [+0.035, +0.054]**	**-0.041 [-0.049, -0.034]**
gpt-4o-mini-verbal	**+0.052 [+0.044, +0.059]**	**+0.034 [+0.026, +0.042]**	**+0.018 [+0.012, +0.023]**	**-0.016 [-0.022, -0.011]**
deepseek-v4-flash-verbal	**+0.023 [+0.017, +0.029]**	**+0.012 [+0.006, +0.017]**	**+0.011 [+0.006, +0.016]**	**-0.011 [-0.014, -0.007]**
gemini-3.8-flash	**-0.012 [-0.018, -0.007]**	-0.001 [-0.004, +0.002]	**-0.012 [-0.017, -0.006]**	-0.001 [-0.003, +0.000]

E1 preregistered per-task criterion (Holm across 8 tasks within model)

Model	Provider	Tasks	shift better (Holm)	told better (Holm)	Mean Δ Brier
jev	TypeSafe	8	3	0	+0.0097

qwen3.7-flash	Alibaba	8	1	0	+0.0315
deepseek-v4-flash	DeepSeek (StreamLake)	8	4	0	+0.0467
gpt-4o-mini	OpenAI	8	5	0	+0.1023
llama-3.3-70b	Meta (Parasail)	8	6	0	+0.2320
gpt-4o-mini-verbal	OpenAI	8	2	0	+0.0341
deepseek-v4-flash-verbal	DeepSeek (StreamLake)	8	3	0	+0.0117
gemini-3.8-flash	Google	8	0	0	-0.0007

Reading the results

- **Mechanism** (exploratory, E1).
  - Telling roughly halves each model's miss of the base rate (mean absolute miss 0.05–0.35 when told, vs 0.11–0.39 zero-shot) but leaves a residual. Llama gets worse when told, because its first-token probabilities are nearly binary.
  - Ranking doesn't change (AUROC told ≈ zero-shot).
  - The code-side shift removes the miss exactly and keeps the ranking.
- **This is not only an artefact of logprob elicitation.** The same two models asked to *state* probabilities show the same sign, with smaller effects.
- **Where the finding stops:** Gemini 3.8 Flash ranks these tasks best (zero-shot AUROC ≈ 0.70). Told Gemini beats the constant, and told ≈ shift: +0.004 in E1-R, significant but tiny.
- **Sobering baseline result (post hoc).** On the leak-repaired tasks, only Gemini's probabilities clearly beat the constant. For the cheap models, even the best code-side fix (Platt on 100 labels) only matches the constant, and LR(100) beats their shifted scores.

Survived vs killed

Survived / new	Killed or weakened
• **NEW (confirmed in E1-R):** telling the base rate scores worse than the constant, for 7/8 configurations from 5 providers. • Telling scores worse than the code shift for 8/8 in E1-R. Gemini's prediction was a null; it came out a tiny positive. • Telling beats zero-shot on most configurations. Telling helps, but not enough. • v1: in-context examples help ranking on 5/8 tasks (not re-run).	• v1 "shift beats telling on 5/8 for Jev" → 3/8 on repaired data. • v1 "one sentence fixes most of it": measured against zero-shot, not the constant. • Shelter, crimes and GitHub v1 AUROCs: inflated by leaks. • All v1 Opus claims: withdrawn (provider-route confound, contamination unchecked). • Full list: CLAIMS.md (12 killed claims).

What's next

1. Finish Haiku (E1 + E1-R), then the final Holm over all 9 configurations. Send the CONFIRMED/NEW message.
2. Red-team the finding:
   - Does it hold on tasks where the model ranks well (AUROC ≥ 0.65)?
   - Is it the sentence's position or wording? E1 already tested 2 extra wordings.
   - Is it the one-token logprob protocol?
3. README rewrite as a neutral guide, DIGEST, website draft.

API spend (v2 ledger; v1 spent $33.56 before this session)

Model	Decisions	Cost
claude-haiku-4.5	15,155	$1.299
llama-3.3-70b	22,900	$0.624
gemini-3.8-flash	23,180	$0.620
gpt-4o-mini	22,764	$0.463
gpt-4o-mini-verbal	30,030	$0.268
deepseek-v4-flash	22,833	$0.146
qwen3.7-flash	22,721	$0.103
jev	23,180	$0.083
deepseek-v4-flash-verbal	23,180	$0.070
Total v2		**$3.68**

Hard cap for this session: $30. The OpenRouter account is shared with other agents.